2303917720521303

Ex.No : 10 Practice with XML DTD

Date :

Aim:

To practice with XML Document Type Definition

Procedure:

Types of DTD:

1)Internal

```
<?xml version="1.0"?>
<!DOCTYPE note [
<!ELEMENT note (to,from,heading,body)>
<!ELEMENT to (#PCDATA)>
<!ELEMENT from (#PCDATA)>
<!ELEMENT heading (#PCDATA)>
<!ELEMENT body (#PCDATA)> ]>
<note>
<to>Tove</to>
<from>Jani</from>
<heading>Reminder</heading>
<body>Don't forget me this weekend</body>
</note>
```

2)External

```
<?xml version="1.0"?>
<!DOCTYPE note SYSTEM "note.dtd">
<note>
  <to>Tove</to>
  <from>Jani</from>
  <heading>Reminder</heading>
  <body>Don't forget me this weekend! </body>
</note>
```

note.dtd

```
<!ELEMENT note (to,from,heading,body)>
<!ELEMENT to (#PCDATA)>
<!ELEMENT from (#PCDATA)>
<!ELEMENT heading (#PCDATA)>
<!ELEMENT body (#PCDATA)>
```

```
<note>
  <to>Tove</to>
  <from>Jani</from>
  <heading>Reminder</heading>
  <body>Don't forget me this weekend</body>
</note>
```

The DTD above is interpreted like this:

- **!DOCTYPE note** defines that the root element of this document is note
- **!ELEMENT note** defines that the note element must contain four elements: "to,from,heading,body"- 4 children, ANY if not fixed.

  <!ELEMENT note ANY>
- **!ELEMENT to** defines the 'to' element to be of type "#PCDATA"
- **!ELEMENT from** defines the 'from' element to be of type "#PCDATA"
- **!ELEMENT heading** defines the 'heading' element to be of type "#PCDATA"
- **!ELEMENT body** defines the 'body' element to be of type "#PCDATA"

PCDATA: Parsed Character Data

CDATA: Character Data

Elements:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE note [
  <!ELEMENT note (to, from, header?, (message+ | body), footer*)>
  <!ELEMENT to (#PCDATA)>
  <!ELEMENT from (#PCDATA)>
  <!ELEMENT header (#PCDATA)>
   <!ELEMENT message (#PCDATA)>
  <!ELEMENT body (#PCDATA)>
  <!ELEMENT footer (#PCDATA)>
]>
<note>
  <to>Alice</to>
  <from>Bob</from>
  <header>Meeting Reminder</header>
  <message>Don't forget the meeting at 10 AM.</message>
  <message>Bring the quarterly report.</message>
  <footer>Sent from my XML client.</footer>
  <footer>Confidential</footer>
</note>
```

1. ‘message’ must occur one or more times.
2. 'to', 'from' are simple text elements.
3. 'header' is optional (0 or 1).
4. 'body' is an alternative to one or more 'message' elements.
5. 'footer' is zero or more times.

Entities

Internal

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE note [
  <!ENTITY company "OpenAI Inc.">
]>
<note>
  This note is from &company;.
</note>
```

External

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE note [
  <!ENTITY company SYSTEM "company.txt">
]>
<note>
  This note is from &company;.
</note>
```

```
<note> This note is from OpenAI Inc.. </note>
```

Attributes

```
<?xml version="1.0" encoding="UTF-8"?>

<!DOCTYPE document [

  <!-- 1. CDATA (with default value) -->

  <!ELEMENT square EMPTY>

  <!ATTLIST square width CDATA "100">

  <!-- 2. Enumerated type (with default value) -->

  <!ELEMENT payment EMPTY>

  <!ATTLIST payment type (cash|check) "cash">

  <!-- 3. ID (must be unique) -->

  <!ELEMENT person EMPTY>
```

```
<!ATTLIST person pid ID #REQUIRED>
<!-- 4. IDREF (must refer to a valid ID) -->
<!ELEMENT employee EMPTY>
<!ATTLIST employee manager IDREF #REQUIRED>
<!-- 5. IDREFS (must refer to multiple IDs) -->
<!ELEMENT team EMPTY>
<!ATTLIST team members IDREFS #REQUIRED>
<!-- 6. NMTOKEN (single valid XML name) -->
<!ELEMENT tag EMPTY>
<!ATTLIST tag name NMTOKEN #REQUIRED>
<!-- 7. NMTOKENS (list of valid XML names) -->
<!ELEMENT labels EMPTY>
<!ATTLIST labels list NMTOKENS #REQUIRED>
<!-- 8. ENTITY (single unparsed external entity) -->
<!NOTATION imgformat SYSTEM "image/jpeg">
<!ENTITY logo SYSTEM "logo.jpg" NDATA imgformat>
<!ELEMENT image EMPTY>
<!ATTLIST image src ENTITY #REQUIRED>
<!-- 9. ENTITIES (list of unparsed external entities) -->
<!ENTITY icon1 SYSTEM "icon1.jpg" NDATA imgformat>
<!ENTITY icon2 SYSTEM "icon2.jpg" NDATA imgformat>
<!ELEMENT icons EMPTY>
<!ATTLIST icons files ENTITIES #REQUIRED>
<!-- 10. NOTATION (must match declared notation) -->
<!ELEMENT media EMPTY>
<!ATTLIST media format NOTATION (imgformat) #REQUIRED>
<!-- 11. xml:lang (predefined, no DTD declaration needed) -->
<!ELEMENT paragraph (#PCDATA)>
<!-- 12. #IMPLIED attribute -->
```

```
    <!ELEMENT contact EMPTY>
    <!ATTLIST contact phone CDATA #IMPLIED>
    <!-- 13. #FIXED attribute -->
    <!ELEMENT sender EMPTY>
    <!ATTLIST sender company CDATA #FIXED "Microsoft">

]>
<document>
    <!-- 1. CDATA with default (overridden here) -->
    <square width="200" />
    <square />
    <!-- 2. Enumerated (check or cash) -->
    <payment type="check" />
    <payment />
    <!-- 3. ID -->
    <person pid="p1" />
    <person pid="e1" />
    <person pid="e2" />
    <!-- 4. IDREF -->
    <employee manager="p1" />
    <!-- 5. IDREFS -->
    <team members="e1 e2" />
    <!-- 6. NMTOKEN -->
    <tag name="Token_123" />
    <!-- 7. NMTOKENS -->
    <labels list="Label1 Label_2 Label3" />
    <!-- 8. ENTITY -->
    <image src="logo" />
    <!-- 9. ENTITIES -->
```

```
    <icons files="icon1 icon2" />
    <!-- 10. NOTATION -->
    <media format="imgformat" />
    <!-- 11. xml:lang -->
    <paragraph xml:lang="en">This is a paragraph in English.</paragraph>
    <!-- 12. #IMPLIED -->
    <contact phone="555-1234" />
    <contact />
    <!-- 13. #FIXED -->
    <sender company="Microsoft" />

</document>
```

Result:

Thus, XML DTD was practiced successfully.